Highly symmetric hypertopes: distinguished groups and finite constructions

Human Compression project

Abstract

We assemble the general and finite results of Fernandes–Leemans–Weiss in their fixed accepted manuscript. Residue stabilizers supply the C-group and C^+-group assertions; an explicit chamber reflection supplies the converse under its geometric hypotheses. Complete certificates identify all twenty-three finite table entries and the three examples. The synthesis retains the classical string and toroidal classification inputs, corrects the finite captions and one group name, and separates the source's open questions. The two complete component manuscripts and their certificates are part of this proof, rather than replaced by the present summary.

1 The principal statements and their hypotheses

A hypertope is a thin, residually connected incidence geometry of finite rank. Every flag extends to a chamber. Automorphisms in this paper preserve types; correlations that permute types are not included in Aut_I. A regular hypertope has a chamber-transitive automorphism group. A chiral hypertope has exactly two chamber orbits, with adjacent chambers in different orbits. Propagation through residues makes both actions free.

Write $I = \{0, \dots, r-1\}$. For involutions ρ_i put $G_S = \langle \rho_i : i \in S \rangle$. A C-group satisfies $G_S \cap G_T = G_{S \cap T}$ for every $S, T \subseteq I$. For generators $\alpha_1, \dots, \alpha_{r-1}$ put $\alpha_0 = 1$ and

$$H_S = \langle \alpha_i^{-1} \alpha_j : i, j \in S \rangle.$$

The C^+ condition is $H_S \cap H_T = H_{S \cap T}$ for $|S|, |T| \geq 2$, with trivial groups on singleton and empty supports.

Theorem 1. *The principal results in the fixed source hold with the following precise conventions.*

1. *A regular hypertope has distinguished nonidentity involutions ρ_i forming a C-group. Conversely, the standard parabolic coset system of a C-group is a regular hypertope if the designated group is flag-transitive. Its full type-preserving automorphism group is then that group.*

2. *A chiral hypertope of rank at least three has independent distinguished generators α_i forming a C^+-group. Its group is transitive on flags of every proper type, so every proper truncation is flag-transitive; thinness of a truncation is not asserted.*

3. *Let a C^+-group G define the standard coset system with $P_i = H_{I \setminus \{i\}}$. Assume that this system is a hypertope and that G has exactly two chamber orbits. It is chiral, with full group G, if and only if no automorphism of G simultaneously inverts all the α_i. Otherwise its full group is $G \rtimes \langle F \rangle$, where F is a geometric involution interchanging the chamber orbits. If θ is the inverting group automorphism, $FgF^{-1} = \theta(g)$.*

4. *At each of the twelve parameters of source Table 1 and eleven parameters of source Table 2, the specified presentations give the finite regular or chiral objects and full group orders in Table 1 below, with the corrected group identification stated there.*

Proof. We acquire the complete general proof [2] and finite proof [3], together with their distinct reviews. The essential interfaces are as follows. Fix a chamber $C = \{c_i : i \in I\}$ and let $F_{I\setminus S}$ be its complementary flag. With left actions, $C^{ij} = (C^i)^j$, the general proof establishes the identities

$$G_S = \mathrm{Stab}_G(F_{I\setminus S}), \qquad H_S = \mathrm{Stab}_G(F_{I\setminus S})$$

in the regular and chiral cases respectively. In the second identity, $\alpha_i C = C^{0i}$ and $\alpha_i^{-1}\alpha_j C = C^{ij}$. Actual paths within the residue prove the first identity, and even paths prove the second; neither proof assumes an unidentified residue group. Intersections of the stabilizers give both intersection conditions and the independence assertion. The standard C-group converse uses flag transitivity to identify every flag and its residue, and then the original intersection property proves thinness and residue connectivity.

For the third assertion, the same proof identifies *every* chamber as $(g,+) = gC$ or $(g,-) = gC^0$, with

$$(g,+) \sim_i (g\alpha_i^{-1},-).$$

The acquired two-orbit theorem gives the reflection $(g,\pm) \mapsto (\theta(g),\mp)$. Reconstruction from colored chamber adjacencies makes it an incidence automorphism. This is a geometric involution even if θ is the identity. An abstract semidirect product, without this reconstruction, would not pay the converse.

The finite component checks the hypotheses just used for all twenty-three presentations: their exact finite groups, all marked intersections, actual flag completions and residue graphs, and all-relator inversion tests. Its regular completions and named-group maps are explicit isomorphisms. For Table 1 its forty-eight subgroup-product inclusions additionally prove that incidence ordered by type is transitive, giving the full polytope axioms. Its computations use right cosets; the bijection $P_i g \mapsto g^{-1}P_i$ preserves types and incidence and converts them to the left-coset convention of the general proof. Thus these are the same geometries. No new finite search is required for this join. □

2 The finite rows and source corrections

The source's printed g is the order of the full type-preserving group. It is the number of chambers in a regular row, and *half* that number in a chiral row. Thus both captions require the same correction.

p	(a,b)	Table 1 group order	Table 2 group order	status
3	(2,0)	240	120	regular
3	(3,0)	1296	648	regular
3	(4,0)	15360	7680	regular
3	(1,2)	336	168	chiral
3	(1,3)	2184	1092	chiral
3	(1,4)	8064	4032	chiral
3	(2,2)	2880	1440	regular
3	(2,3)	6840	3420	chiral
4	(1,1)	288	—	regular
4	(2,0)	768	384	regular
4	(1,2)	2016	1008	chiral
5	(2,0)	28800	14400	regular

Table 1. Exact positive finite roster; no completeness claim for chiral rows.

The Table 2 group at $(3,2,2)$ is

$$D = \{(s,t) \in S_5 \times S_4 : \operatorname{sgn}(s) = \operatorname{sgn}(t)\},$$

not $A_5 \times S_4$. The finite proof gives a bijective map to D and distinguishes the two groups by the centralizers of their characteristic perfect residuals, of orders 12 and 24. It identifies the groups abbreviated $[1\,1\,2]^3$ and $[1\,1\,2]^4$ by explicit $G(3,3,4)$ and G_{29} models. Every other printed group name is likewise identified by an actual map, including the actions in the split extensions. Each of the eleven Table 2 full groups embeds with index two in its corresponding Table 1 group; the complete marked halving maps and complements are retained.

The three examples are also acquired in full. Example 4.4 has group $(C_3 \times C_3) \rtimes C_2$ and incidence graph $K_{3,3,3}$, with 27 chambers and full group S_3^3. Its designated order-18 group is not flag-transitive and its panels have three completions. Example 4.5 has the printed C-marking of A_6, 720 chambers, 360 panels with four completions and 36 disconnected residues. Example 6.3 has order 72, satisfies the involutory intersection property, and has rotation group equal to the whole group; two rotation subgroups intersect in order 8 instead of the required 4. These are separate reasons why the relevant geometric or index-two hypotheses cannot be discarded.

One additional source assertion about Example 4.5 needs qualification.

Proposition 2. *The rotation marking of Example 4.5 is not a C^+-marking. Consequently the sentence on source page 17 does not supply its claimed C^+ example.*

Proof. Retain the source involutions $\rho_0, \rho_1, \rho_2, \rho_3$ and set $\alpha_i = \rho_0\rho_i$. The acquired finite certificate gives $G = A_6$, the two parabolics $G_{\{0,2,3\}}$, $G_{\{0,1,3\}}$ of order 60, and their intersection $G_{\{0,3\}}$ of order 10. Every index-six subgroup of A_6 is isomorphic to A_5: the action on its six cosets is faithful by simplicity, has image A_6 in S_6, and identifies it with a point stabilizer. For a group generated by involutions, the subgroup generated by their pairwise products is normal and has index at most two. Thus simplicity of A_5 gives

$$H_{\{0,2,3\}} = G_{\{0,2,3\}}, \qquad H_{\{0,1,3\}} = G_{\{0,1,3\}}.$$

Their intersection has order 10, whereas $H_{\{0,3\}} = \langle \rho_0\rho_3 \rangle$ has order 5. Equivalently ρ_0 lies in the former intersection and not the latter. This violates IC^+. The order and group-identification data are already part of Example 4.5's accepted certificate; there is no additional census. □

This corrects the later illustration, not the local group and incidence statements of Example 4.5. In particular it neither proves nor refutes a general converse for arbitrary C^+-groups. Theorem 1(3) keeps its hypertope and exactly-two-orbit hypotheses throughout.

3 Complete source routing and retained inputs

The fixed source is the twenty-two-page accepted author manuscript of [1]. The complete route register accompanies this paper. The nineteen named general clauses are routed as follows:

Source clauses	Complete acquired route
2.1–2.2	Propagation within every residue and flag extension.
3.1–3.5	Cosets and common-core quotient; actual chamber and panel stabilizers; Dehon and Buekenhout–Hermand criteria.
4.1–4.3, 4.6	Regular residue identity, Tits' three-coset criterion, rank-three panel completions and the flag-transitive converse.
5.1–5.2	Retained marked string correspondence and the proved rank-two residue interpretation of a string diagram.
6.1–6.2	Independence for rotations and the even-word kernel at index two.
7.1–7.2	Even residue paths, full support intersections and independence.
8.1–8.2	Proper-flag transitivity and exact two-orbit coordinates.

Construction 8.1 defines an incidence system; flag extension and hypertope properties are not automatic. A coset action is faithful only after factoring its common core. In a nonfaithful chamber-transitive hypertope action the Borel subgroup equals that core, rather than necessarily being trivial. The proof of source Proposition 2.1 uses connectedness of each residue in its induction, not connectedness of the whole incidence graph alone. These qualifications are explicit in the general component.

Four previously established inputs remain attributed premises, rather than new classification proofs: the marked string C-group correspondence (Aschbacher, Schulte and McMullen–Schulte); the rank-three toroidal chirality criterion of Breda D'Azevedo–Nedela; the toroidal hypermap families of Corn–Singerman; and McMullen–Schulte's finite regular locally toroidal polytope classification at $p = 3, 4, 5$. The toroidal inputs retain their admissible, nondegenerate parameter domains; the unrestricted wording on source pages 12–13 cannot include the unit period $(1, 0)$. Indeed its displayed rotation presentation gives $y = x^2$ and the cyclic group C_3, with coincident parabolics and no thin geometry. The exceptional $(1, 1)$ system is Example 4.4. We do not assert a larger parameter domain by extrapolating the finite tables. The positive rows above are verified independently of classification completeness.

Earlier attributed remarks about polytope dualities, the 11-cell, Schulte–Weiss rotations, chiral extensions, codimension-two faces and the Coxeter honeycomb construction remain contextual antecedents. They are not new classifications, additional finite table rows or hypotheses secretly used in Theorem 1. The $p = 6$ programme is explicitly deferred by this source; this synthesis does not turn that announcement into a new L052 obligation or a fresh intake.

4 Questions, evidence and attribution

Problems 9.1 and 9.2 and the unnumbered completeness conjecture for the known chiral list remain questions. Checking the positive rows does not prove the latter two classifications. The programme's earlier negative answers to Problems 9.3 and 9.4 are acquired with their precise standard parabolic hypotheses: in 9.3 thinness and chamber connectedness force the designated chamber orbit to be all chambers; in 9.4 the rank-three standard construction is residually connected. The latter uses a connected spanning coset graph, not an unjustified equality of incidence relations. These old answers receive no new question credit and retain their previous limited formalization scope.

The mathematical results and classical inputs keep the attributions in [1]. The general component was authored by the root collaborator and independently reviewed by the present assembler. The finite component was authored by a separate collaborator and independently reviewed by root. The present contribution is the complete source join and Proposition 2; the finite author confirmed its convention without supplying a new finite computation. The accepted chamber and L003 shared proofs remain complete acquired dependencies. Named group constructors, exact presentation/isomorphism routines and the named CHEVIE model are explicit software and identification premises of the finite certificate. No new Lean theorem, measurement, compression factor or priority claim is made. Whole-source admission still requires a distinct review of this assembly and the publication gate.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Math. 90 (2016), 1045–1067; fixed accepted author manuscript, 22 pages.

[2] *Residue stabilizers and distinguished generators*, accepted complete general recovery, eight pages; exact author, independent review and root-adoption editions acquired.

[3] *Finite coset geometries in Highly symmetric hypertopes*, accepted complete finite recovery, five pages; exact author and distinct root-review editions acquired.

[4] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002. Marked correspondence and the cited regular locally toroidal classifications retained.

[5] A. Breda D'Azevedo and R. Nedela, *Chiral hypermaps of small genus*, Beitr. Algebra Geom. 44 (2003), 127–143. Retained source input.

[6] D. Corn and D. Singerman, *Regular Hypermaps*, J. Combinatorics 9 (1988), 337–351, as cited in the fixed source.

Residue stabilizers and distinguished generators

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This manuscript supplies the general arguments for the intake of Fernandes–Leemans–Weiss, *Highly symmetric hypertopes* (2016). Its central observation is that a stabilizer of a flag can be read from paths of chambers in its residue. Ordinary paths give the distinguished involutions of a regular hypertope; even paths give the distinguished generators of a chiral hypertope. Intersections then follow from intersections of pointwise stabilizers. We use the already acquired propagation, extension-fiber and reflection theorems in the chamber manuscript. Their proofs are dependencies, not new assertions of the present application. Finite examples, numerical tables and the toroidal families are separate components.

1 Conventions and acquired geometric arguments

Let I be a finite type set. An incidence geometry has one object of each type in every chamber, and every flag extends to a chamber. Distinct objects of the same type are not incident. The residue of a flag F consists of the objects of the remaining types incident with every object of F. Residual connectedness means incidence connectedness of every residue of rank at least two, including the empty-flag residue. Thinness means that every panel is contained in exactly two chambers.

For a thin geometry write C^i for the unique chamber different from C on the panel obtained by deleting its type-i object. Superscripts are applied in their written order: $C^{ij} = (C^i)^j$. All group actions in this manuscript are *left* actions; thus $(gh)C = g(hC)$. An automorphism preserves types unless stated otherwise. It commutes with every operation $C \mapsto C^i$.

The acquired propagation and reconstruction lemma has three conclusions:

1. Chambers containing a fixed flag are connected by adjacency steps which retain that flag.
2. In a thin residually connected geometry, an automorphism fixing a chamber is the identity.
3. A permutation of chambers preserving every adjacency color comes from a unique automorphism of the incidence geometry.

The proof of the first conclusion is an induction on the rank of the residue. It is important to apply the induction to residues known to be connected. Mere incidence connectedness of the whole geometry does not supply that hypothesis. The converse needed for Proposition 2.1 of the source is elementary: a chamber path in a residue of rank at least two gives an incidence path between any two of its objects. Consecutive chambers share at least one object, and each object lies in a chamber. Thus residual connectedness and strong chamber connectedness are equivalent under the source's hypotheses. This corrects an intermediate step in its proof without changing the stated proposition.

Chamber transitivity implies transitivity on flags of every fixed type: extend both flags to chambers and carry one chamber to the other. Conversely chamber transitivity is a special case of flag transitivity. This is Proposition 2.2 of the source.

2 The residue stabilizer identity

Fix a chamber $C = \{c_i : i \in I\}$ and put

$$F_{I\setminus S} = \{c_i : i \notin S\} \qquad (S \subseteq I).$$

Regularity means transitivity of the full automorphism group on chambers; by propagation its action is free as well. The same conclusions below hold for any chamber-transitive group of actual automorphisms, which then equals the full group by freeness.

Theorem 2.1. *Let G act regularly on the chambers of a hypertope. There is a unique $\rho_i \in G$ with $\rho_i C = C^i$ for each $i \in I$. These elements are distinct nonidentity involutions. For every $S \subseteq I$,*

$$\langle \rho_i : i \in S\rangle = \mathrm{Stab}_G(F_{I\setminus S}). \tag{1}$$

In particular they generate G and satisfy the full intersection property

$$\langle \rho_i : i \in S\rangle \cap \langle \rho_i : i \in T\rangle = \langle \rho_i : i \in S \cap T\rangle.$$

Proof. Existence and uniqueness of ρ_i follow from regularity. Since $\rho_i(C^i) = (\rho_i C)^i = C$, its square fixes C, so $\rho_i^2 = 1$. It is nonidentity because $C^i \neq C$. The chambers C^i, C^j are distinct when $i \neq j$, so the generators are distinct. Each ρ_i fixes every c_j with $j \neq i$. This proves inclusion from left to right in (1).

Let g fix $F_{I\setminus S}$. The chambers C and gC contain this flag. Propagation inside its residue gives a path from C to gC whose colors all belong to S. If its colors are $i_1, \ldots, i_m$, its successive chambers are

$$C, \quad \rho_{i_1}C, \quad \rho_{i_1}\rho_{i_2}C, \quad \ldots, \quad \rho_{i_1}\cdots\rho_{i_m}C.$$

Indeed adjacency commutes with every group element. The last product and g have the same image of C, and freeness makes them equal. This also works for $S = \varnothing$, when the stabilizer is trivial. It proves (1). Taking $S = I$ proves generation. Finally, to fix both complementary flags is exactly to fix their union, which is the complementary flag of $S \cap T$. Applying the identity three times gives the claimed intersection. □

This proves source Theorem 4.1 without an induction that presumes the identification of the residue group. Its rank-two Coxeter quotient interpretation follows as well: put $m_{ij} = o(\rho_i\rho_j)$, allowing ∞. The Coxeter presentation with these labels maps onto G. The rank-two subgroups are dihedral of order $2m_{ij}$ (infinite when the label is infinite), so the map is injective on each such subgroup.

For the chiral case, $G = \mathrm{Aut}_I(\Gamma)$ has exactly two chamber orbits, and adjacent chambers lie in different orbits. Call the orbit of C positive. Every two-step chamber C^{ij} is positive. The action is free by propagation, so there is a unique element taking C to C^{ij}. Fix $0 \in I$, put $\alpha_0 = 1$, and for $i \neq 0$ define $\alpha_i C = C^{0i}$.

Theorem 2.2. *For a chiral hypertope of rank at least three set*

$$\alpha_{ij} = \alpha_i^{-1}\alpha_j, \qquad H_S = \langle \alpha_{ij} : i, j \in S\rangle \quad (S \subseteq I).$$

Then $\alpha_{ij}C = C^{ij}$ and

$$H_S = \mathrm{Stab}_G(F_{I\setminus S}). \tag{2}$$

Consequently $G = \langle \alpha_i : i \neq 0\rangle$ and $H_S \cap H_T = H_{S\cap T}$ for all subsets S, T. The set of nonidentity distinguished generators $\{\alpha_i : i \neq 0\}$ is independent.

Proof. First $\alpha_i^{-1}C = C^{i0}$, since applying α_i to the latter chamber gives $C^{0ii0} = C$. Therefore

$$\alpha_i^{-1}\alpha_j C = \alpha_i^{-1}(C^{0j}) = (C^{i0})^{0j} = C^{ij}.$$

The same identity includes $i = 0$, $j = 0$ and $i = j$. In particular α_{ij} fixes every object of C outside types i, j, so H_S is contained in the stabilizer in (2).

For the reverse inclusion let g fix the complementary flag. Propagation gives a path from C to gC using colors in S. Every adjacency changes the chamber orbit. Since both endpoints are positive, the path has even length, say with colors $i_1, j_1, \ldots, i_m, j_m$. After its first pair the chamber is $\alpha_{i_1 j_1} C$. Equivariance then shows successively that its last chamber is $\alpha_{i_1 j_1} \cdots \alpha_{i_m j_m} C$. Freeness makes this product equal to g.

This argument includes singleton and empty S. Their group H_S is trivial; equivalently a group element stabilizing a panel cannot exchange its two chambers because those chambers have different colors. We have proved the stabilizer identity for every S, rather than assumed that a residue has at most two orbits. Generation follows on taking $S = I$; intersections follow by taking the union of the two fixed flags.

The chamber C^{0i} differs from C for $i \neq 0$: equality would imply $C^0 = C^i$, which is impossible for different types. Hence $\alpha_i \neq 1$. The subgroup generated by the other α_j is $H_{I\setminus\{i\}}$, whereas $\alpha_i \in H_{\{0,i\}}$. Their intersection is $H_{\{0\}} = 1$. Thus no α_i lies in the subgroup generated by the others. □

The intersection conclusion contains the source's condition IC^+, which is stated only for $|S|, |T| \geq 2$. This proves Theorem 7.1 and Corollary 7.2. Ranks at most two have no chiral hypertopes: a connected thin rank-two incidence graph is a cycle or a doubly infinite path, whose type-preserving automorphisms are transitive on its edges; the rank-one case is the two-object geometry.

Corollary 2.3. *The chiral group is transitive on flags of every proper type set. In particular every proper truncation is flag-transitive.*

Proof. Let F be a flag of proper type J and extend it to a chamber D. If D is negative, choose $i \notin J$ and replace it by D^i; this retains F and makes the chamber positive. Some $g \in G$ takes C to the resulting positive chamber and therefore takes its J-subflag to F. A flag in a truncation extends to a chamber of that truncation by extending it in the whole geometry. The same transitivity argument applies there. □

This includes Proposition 8.1 of the source. It asserts transitivity, not thinness or regularity of those truncations.

3 Cosets and the converse construction

For subgroups $P_i \leq G$, the objects of type i are left cosets gP_i, with incidence given by nonempty intersection. This always defines an incidence system and has the base chamber $\{P_i : i \in I\}$. It need not have the flag-extension property in rank at least four. Left multiplication is transitive on objects of each type and on incident pairs: take a common representative of the two cosets. These are the precise assertions used from source Proposition 3.1.

The kernel of the action is

$$K = \bigcap_{i \in I} \bigcap_{g \in G} g P_i g^{-1}.$$

It is contained in every P_i. Sending gP_i to $(gK)(P_i/K)$ is a bijection on each type. It preserves incidence in both directions, because a representative of an intersecting quotient coset can be lifted and the discrepancy is in $K \leq P_i$. Thus it gives the isomorphism of source Theorem 3.3. A claim that G itself acts faithfully before this quotient would be false.

If a group is transitive on objects and incident pairs of each type, its geometry is reconstructed by these cosets of the stabilizers of a base chamber. Indeed $gP_i \mapsto gc_i$ is a well-defined bijection; two image objects are incident precisely when the pair is a translate of the base pair, which is equivalent to the two cosets intersecting. This applies to regular hypertopes and, by Corollary 2.3, to chiral hypertopes of rank at least three. The residue stabilizer identities identify their P_i with the prescribed marked subgroups.

For a regular faithful action the Borel subgroup $B = \bigcap_i P_i$ is the chamber stabilizer and hence is trivial. The stabilizer of a panel has exactly two elements, since it acts transitively and freely on the two containing chambers. These observations prove Lemma 3.4 and its stated consequence. For a nonfaithful chamber-transitive coset action on a hypertope, the corresponding conclusion is $B = K$ and a panel stabilizer has quotient of order two by K; passage to the faithful quotient is essential.

Theorem 3.1. *Let $(G, \{\rho_i : i \in I\})$ be a C-group and put $G_S = \langle \rho_i : i \in S \rangle$ and $P_i = G_{I\setminus\{i\}}$. If its coset incidence system is flag-transitive under G, it is a regular hypertope. Its full automorphism group is G.*

Proof. Every flag is a translate of the corresponding base flag, so extends to a chamber. Iterated use of the C-group intersection property gives

$$\bigcap_{j \notin S} P_j = G_S \qquad (S \subseteq I),$$

including the conventions for empty intersections. The base chamber stabilizer is $G_\varnothing = 1$, so the action is faithful and free on chambers. The stabilizer of the complementary base flag acts transitively on the chambers containing that flag: a group element taking the base chamber to any such chamber must fix every object of the flag. For a panel its stabilizer is $G_{\{i\}} = \{1, \rho_i\}$; hence the geometry is thin.

More generally its residue chambers are the chambers gC with $g \in G_S$. A word in the ρ_i, $i \in S$, gives an adjacency path between C and gC wholly inside that residue. Thus all residues are chamber-connected and therefore incidence-connected in rank at least two. The geometry is a hypertope. Its full automorphism group acts freely on chambers, whereas G already acts transitively, so the two groups coincide. □

This proves source Theorem 4.6. It does not assert flag transitivity from the intersection property alone.

4 The classical coset criteria

The following elementary arguments record exactly which group conditions are used. Their origins remain Tits, Dehon and Buekenhout–Hermand as attributed in the source; they are not new claims of priority.

Proposition 4.1. *Suppose the coset incidence system is flag-transitive. Put $P_J = \bigcap_{j\in J} P_j$, with $P_\varnothing = G$. It is residually connected if and only if*

$$P_J = \langle P_{J\cup\{k\}} : k \in I \setminus J \rangle \qquad (|I \setminus J| \geq 2).$$

Proof. Fix the base flag of type J and write $H = P_J$. Every incident object of type $k \notin J$ has the form hP_k with $h \in H$, by flag transitivity on $J \cup \{k\}$. Two such representatives give the same object precisely when they give the same left coset of $H \cap P_k$ in H. If two objects of the residue are incident, flag transitivity on their union with the base flag supplies a common representative in H. Consequently the residue is exactly the coset incidence system of H with subgroups $H \cap P_k$.

In a coset incidence graph of rank at least two, the component containing the base flag is invariant under each of its vertex stabilizers: an automorphism fixing a vertex preserves that vertex's component. It therefore contains every vertex represented by an element of the group L generated by those stabilizers. Conversely, since each stabilizer is contained in L, assigning the label hL to a vertex represented by h is well-defined and constant along edges. The graph is connected if and only if $L = H$. Apply this to every J. □

Without flag transitivity, a residue can have additional incidence edges; the equality with a coset graph must not be assumed.

Proposition 4.2. *For each nonempty $J \subseteq I$ choose $a(J) \in J$. The coset system is flag-transitive if and only if, for every $|J| \geq 3$,*

$$\bigcap_{j \in J \setminus \{a(J)\}} P_j P_{a(J)} = \left(\bigcap_{j \in J \setminus \{a(J)\}} P_j \right) P_{a(J)}.$$

Proof. The acquired extension-fiber theorem identifies orbits on flags of type $T \cup \{k\}$, assuming transitivity on type T, with

$$P_T \backslash \left(\bigcap_{j \in T} P_j P_k \right) / P_k.$$

There is one orbit precisely when the middle set is $P_T P_k$. Transitivity on flags of size at most two always holds. Inducting on $|J|$, with $T = J \setminus \{a(J)\}$, proves sufficiency. Necessity is the same orbit calculation at every step. Every flag then translates to a base flag and extends, so no missing geometry hypothesis is hidden in this test. □

Lemma 4.3. *For subgroups A, B, D the following are equivalent:*

$$AB \cap AD = A(B \cap D), \tag{i}$$

$$(A \cap B)(A \cap D) = A \cap BD, \tag{ii}$$

and (iii): three right cosets of A, B, D that meet pairwise have a common point.

Proof. Suppose (i), and let $x = bd \in A \cap BD$. Then $b = xd^{-1}$ belongs to $AB \cap AD$, so $b = ae$ with $a \in A$ and $e \in B \cap D$. Since $b, e \in B$, we have $a \in A \cap B$; and $ed = a^{-1}x$ belongs to $A \cap D$. This proves the nontrivial inclusion of (ii).

Conversely let $x = ab = cd \in AB \cap AD$, with $a, c \in A$, $b \in B$, $d \in D$. Inverting equality (ii) also gives $A \cap DB = (A \cap D)(A \cap B)$. Thus $c^{-1}a = db^{-1} = uv$, where $u \in A \cap D$ and $v \in A \cap B$. It follows that $x = cu(vb)$, with $vb = u^{-1}d \in B \cap D$. This gives (i); the opposite inclusions in both equalities are automatic.

For the coset assertion, translate a common point of the B- and D-cosets to the identity by right multiplication. The three cosets then have the form Ak, B, D. They meet pairwise precisely when $k \in AB \cap AD$. They have a common point precisely when $k \in A(B \cap D)$. This is exactly (i). □

This is source Lemma 4.2. Inverting cosets gives the equivalent statement for the left-coset convention used elsewhere here.

Corollary 4.4. *For a rank-three C-group the standard coset geometry is thin if and only if the designated group acts regularly. In either case it is a regular hypertope.*

Proof. Any rank-three coset system is a geometry: an incident pair has a common representative, which supplies an object of the third type. If the group is chamber-transitive, Theorem 3.1 applies.

Now suppose it is thin. Normalize any incident pair to P_i, P_j, and let k be the remaining type. There are two distinct completions P_k and $\rho_k P_k$: the element ρ_k belongs to $P_i \cap P_j$ and not to P_k, by the intersection property. Both completed chambers are translates of the base chamber. Thinness allows no other completion. Hence every chamber is a translate of the base chamber, and the theorem applies. No inference of freeness on all chambers from the base stabilizer alone is used. □

5 Rotation subgroups and strings

Proposition 5.1. *Let $(G, \{\rho_i : i \in I\})$ be a C-group with $0 \in I$. Set $\alpha_0 = 1$, $\alpha_i = \rho_0\rho_i$ for $i \neq 0$ and $H = \langle \alpha_i : i \neq 0 \rangle$. These generators are independent. The index $[G : H]$ is one or two. If it is two, then*

$$\langle \alpha_i^{-1}\alpha_j : i, j \in S \rangle = G_S \cap H \qquad (S \subseteq I),$$

so the rotation generators satisfy IC^+.

Proof. Conjugation by ρ_0 inverts every α_i, and $\rho_i = \rho_0\alpha_i$. Thus $H \lhd G$ and G/H is generated by the image of ρ_0, whose order is at most two. If α_i lay in the subgroup generated by the other α_j, it would belong to $G_{I\setminus\{i\}}$. Multiplication by ρ_0 would put ρ_i in that subgroup, contradicting the intersection property. This proves independence regardless of the index.

If the index is two, all ρ_i have the same nontrivial image in G/H. Consequently a word in the ρ_i, $i \in S$, represents an element of H exactly when its length is even. Every such word is a product of pairs $\rho_i\rho_j = \alpha_i^{-1}\alpha_j$ with $i, j \in S$. This proves the displayed identity. Intersecting two such identities and using $G_S \cap G_T = G_{S\cap T}$ proves IC^+. □

These are Propositions 6.1 and 6.2. The index-two hypothesis in the last assertion cannot be discarded; the source's Example 6.3 is checked in the finite component.

For source Theorem 5.1 we retain the standard string C-group correspondence with abstract regular polytopes, in its exact marked form: the faces of type i are the cosets of $G_{I\setminus\{i\}}$, its flag action is regular, and its rank-two sections have polygon sizes $o(\rho_i\rho_j)$. This is the classical Aschbacher–Schulte input cited in that theorem and the McMullen–Schulte correspondence already retained by the programme. It yields flag transitivity, after which Theorem 3.1 gives the hypertope statement. This input is not counted as a new proof here.

Conversely, for a regular hypertope the rank-two residue on types i, j has group $G_{\{i,j\}}$ by Theorem 2.1. Its alternating chamber cycle has length $2o(\rho_i\rho_j)$, or is infinite if that order is infinite. A missing edge in its diagram means this residue is a digon, so the product order is two. If the diagram is a string, every pair of nonconsecutive types is missing, and the distinguished C-group is therefore a string C-group. This proves source Theorem 5.2.

6 The two chamber orbits in Construction 8.1

Let $I = \{0, \ldots, r-1\}$ with $r \geq 3$. Suppose $G = \langle \alpha_1, \ldots, \alpha_{r-1} \rangle$ is a C^+-group with its independent nonidentity generators. Set $\alpha_0 = 1$ and $H_S = \langle \alpha_i^{-1}\alpha_j : i, j \in S \rangle$. The parabolics of Construction 8.1 are

$$P_i = H_{I\setminus\{i\}} \quad (i \in I).$$

For $i > 0$ this is $\langle \alpha_j : j \neq 0, i \rangle$; for $i = 0$ it is $\langle \alpha_1^{-1}\alpha_j : 2 \leq j < r \rangle$, by telescoping the pairwise differences. We continue to assume, as the source theorem does, that the associated coset system is a hypertope and that G has *exactly two* orbits on its chambers.

Iterating IC^+ gives trivial intersections of all parabolics but one: the common support is a singleton and its H-group is trivial. The iteration uses IC^+ while both supports have size at least two; as soon as a singleton is reached the intersection is already trivial. The base chamber stabilizer is trivial as well. Thus this is a faithful action, and propagation makes it free on every chamber.

Write $C = \{P_i : i \in I\}$. The coset $\alpha_1 P_0$ contains every α_j, $j > 0$. For each $i > 0$ choose $j > 0$ different from i; then $\alpha_j \in P_i$, so replacing P_0 by $\alpha_1 P_0$ gives a chamber. It differs from C: otherwise $\alpha_1 \in H_{I\setminus\{0\}} \cap H_{\{0,1\}} = H_{\{1\}} = 1$. By thinness it is C^0; denote it by C_0.

For $i > 0$, the coset $\alpha_i^{-1} P_i$ is different from P_i by independence. It meets P_k for $k > 0$, $k \neq i$, at α_i^{-1}. It meets P_0 at $\alpha_i^{-1}\alpha_j$, where $j > 0$, $j \neq i$. Replacing P_i by this coset therefore gives the chamber C^i. On multiplying it by α_i, every positive-type object becomes the corresponding object of C_0, while $\alpha_i P_0 = \alpha_1 P_0$. Hence

$$C^i = \alpha_i^{-1} C_0 \qquad (i \in I).$$

Here $i = 0$ follows from $\alpha_0 = 1$.

No element of G takes C to any C^i, since it would fix the base panel and its stabilizer is trivial. The same holds at any translate of C. Also every neighbor of a translate of C_0 is a translate of C: this follows from $C^i = \alpha_i^{-1} C_0$ and involutivity of adjacency. With the exactly-two-orbits hypothesis, all chambers are therefore uniquely

$$(g, +) = gC, \qquad (g, -) = gC_0, \qquad (g, +) \sim_i (g\alpha_i^{-1}, -).$$

These are precisely the hypotheses and coordinates of the acquired two-regular-orbits theorem. It follows that the full group is G, and the hypertope is chiral, unless some $\theta \in \mathrm{Aut}(G)$ inverts every α_i. If such a θ exists, the explicit color-preserving map

$$(g, +) \mapsto (\theta(g), -), \qquad (g, -) \mapsto (\theta(g), +)$$

extends to an incidence involution F, and the full group is $G \rtimes \langle F \rangle$, with $FgF^{-1} = \theta(g)$. This proves source Theorem 8.2 by the already acquired theorem. In this notation the adjoined chamber involution can be nontrivial even when the abstract automorphism θ is the identity.

The proof uses the two-orbits hypothesis to identify the entire chamber set. It does not establish that hypothesis for an arbitrary C^+ presentation. Nor does an abstract semidirect product alone establish that its extra involution acts on the proposed geometry.

7 Scope, attribution and source questions

Theorems 2.1 and 2.2 replace the two separate intersection arguments by the same residue-stabilizer identity. The converse construction and the rotation argument are then short deductions. The geometric propagation lemma, extension-fiber theorem and explicit reflection theorem are acquired project proofs, with their earlier attributions unchanged. The subgroup criteria remain the classical results attributed by the source to Tits, Dehon and Buekenhout–Hermand. The string correspondence remains an explicit retained theorem, not an inferred consequence of an order calculation. No claim of historical priority for the stabilizer observation is made.

The programme already has reviewed negative answers to source Problems 9.3 and 9.4 under the standard parabolic construction. Their precise statements are retained here. Thinness and chamber connectedness force the standard C-group orbit to contain all chambers, excluding the example requested in 9.3. Every rank-three standard coset system on three generators is residually connected, excluding the example requested in 9.4. The latter argument only needs a connected *spanning* coset graph in each residue and does not assert an equality of residue incidence relations without flag transitivity. These are acquisitions of earlier answers, not two new solutions generated by this intake.

Problems 9.1 and 9.2 and the conjectured completeness of the known chiral list remain separately registered questions. This component supplies no classification of all locally toroidal cases. It grants no whole-original credit before the finite and family components and the final source map are reviewed. Existing Lean declarations retain exactly their previous algebraic and action scope; no new formalization or compression measurement is asserted here.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Mathematicae 90 (2016), 1045–1067. Fixed accepted author manuscript, 22 pages.

[2] Human Compression project, *Chambers and marked covers*, topic-chamber/v2, sections on normalization, propagation and reconstruction, and two regular orbits. The accepted version and review are acquired by their exact artifact bindings.

[3] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Section 2E. The marked regular-polytope correspondence is retained, as in the existing programme.

Finite coset geometries in Highly symmetric hypertopes

Human Compression project

Abstract

We verify the three finite examples and all twenty-three table entries in Fernandes–Leemans–Weiss, using their exact markings and presentations. The common coset criteria are acquired from the earlier recovery of *Hexagonal extensions of toroidal hypermaps*; their application here includes the actual flag and residue tests. All printed orders and regularity assertions hold. One group name and both chiral flag-count captions require correction. The completeness conjecture concerning the chiral list is not used or proved.

1 The groups and the certificate interface

We use the accepted author manuscript of the source [1], with physical page numbers. Its Table 2, on page 20, specifies the presentations

$$P(p,a,b) = \langle x,y,z \mid x^3, y^3, z^p, (x^{-1}z)^2, (y^{-1}z)^2, (x^{-1}y)^3, (xy^{-1}x)^a(xy)^b \rangle. \tag{1}$$

The involutory completion, when it exists, is part of the conclusion and is not imposed in this presentation. Put $a_0 = 1, a_1 = x, a_2 = y, a_3 = z$ and

$$P_J = \langle a_i^{-1}a_j : i,j \in J \rangle, \qquad P_i = P_{\{0,1,2,3\}\setminus\{i\}}.$$

The associated incidence system has right cosets of P_i as its elements. Two elements of different types are incident when their cosets intersect.

For Table 1 we use standard polytope rotations u, v, w. The literal source relation (5.1) gives

$$Q(p,a,b) = \langle u,v,w \mid u^6, v^3, w^p, (uv)^2, (vw)^2, (uvw)^2, (vu^{-2})^a(v^{-1}u^2)^b \rangle. \tag{2}$$

Indeed $u = \rho_0\rho_1$, $v = \rho_1\rho_2$, $w = \rho_2\rho_3$, so $vu^{-2} = \rho_1\rho_2(\rho_1\rho_0)^2$ and $v^{-1}u^2 = \rho_2\rho_1(\rho_0\rho_1)^2$. The pair-difference marking for this group is $(1, u, uv, uvw)$. In the program the three nonidentity entries are the free generators; this is the invertible change $u = X, v = X^{-1}Y, w = Y^{-1}Z$. It is essential to keep this change of marking when checking inversion.

We acquire the following complete finite certificate theorem from [2]. For four maximal rotation parabolics, set $K_J = \bigcap_{j\in J} P_j$. The type-$i$ completions of the base flag of types J are counted by

$$c(i,J) = \#\{g \in T : g \in P_iP_j \ (j \in J)\},$$

where T is a right transversal of $P_i \cap P_{j_0}$ in P_{j_0} for one $j_0 \in J$. We check full pair-difference intersection, the equalities $c(i,J) = [K_J : K_J \cap P_i]$ for $|J| = 2$, and $c(i,J) = 2$ for $|J| = 3$. In addition, the P_i generate P, each P_f is generated by its intersections with the other parabolics, and every rank-two residue is connected. The last test uses its actual incidence graph: two copies of K_J, with h joined to k exactly when $hk^{-1} \in P_iP_j$, for the two remaining types i, j.

These conditions imply a thin residually connected geometry with a free action having exactly two chamber orbits. Thus they check more than marked subgroup intersections. Simultaneous inversion of the three marked generators is tested by evaluating *all* defining relators on their inverses. If the relators hold, the induced endomorphism squares to the identity and is an automorphism. Otherwise a failed relator is a literal obstruction. The acquired two-orbit theorem [3] then determines the full type-preserving automorphism group. In the regular case we also construct $H = P \rtimes \langle t \rangle$, put $\rho_i = ta_i$, and verify all 256 subgroup intersections and all normalized flag-completion tests in H. The acquired regular-completion proof identifies this geometry with the rotation coset geometry.

2 All table entries

Theorem 1. *The presentations* (1) *and* (2)*, at precisely the parameters in Table 1, give the regular or chiral geometries recorded there. The displayed numbers are the orders of their full type-preserving automorphism groups. Each regular geometry has that many chambers; each chiral geometry has twice that many chambers.*

p	(a,b)	Table 1 order	Table 2 order	status
3	(2,0)	240	120	regular
3	(3,0)	1296	648	regular
3	(4,0)	15360	7680	regular
3	(1,2)	336	168	chiral
3	(1,3)	2184	1092	chiral
3	(1,4)	8064	4032	chiral
3	(2,2)	2880	1440	regular
3	(2,3)	6840	3420	chiral
4	(1,1)	288	—	regular
4	(1,2)	2016	1008	chiral
4	(2,0)	768	384	regular
5	(2,0)	28800	14400	regular

Table 1: The exact finite parameters.

Proof. For every row, GAP computes an isomorphism from the finitely presented group to a finite permutation group. This supplies both an upper bound for the presentation and a faithful finite image; an arbitrary finite quotient would not suffice. The complete seven-word relator lists are retained. A separate literal-permutation checker verifies the relators, generating orders and all pair-difference subgroup intersections without calling the presentation routine.

The acquired finite criteria of Section 1 are then checked for every row. There are 256 pair-difference intersections, sixteen completion counts, four parabolic-generation tests and six actual rank-two residue graphs per row. The absent inversion is witnessed by the last relator in each chiral row. Each regular row has the full involutory completion and its independent intersection and flag checks. The proper-flag counts prove flag completion and transitivity on all proper types; the two completions of a panel lie in different free chamber orbits. Consequently the full groups and chamber counts are those asserted. For (2) we verify the polytope assertion directly. For each $i < j < k$, the literal subgroups satisfy $P_j \subseteq P_iP_k$. To use this condition, normalize a middle face to P_j. Every incident face of type i or k has a representative in P_j, so write the two faces as P_ih and P_kt with $h, t \in P_j$. Since $ht^{-1} \in P_j \subseteq P_iP_k$, the two faces are incident. Thus incidence, ordered by type, is transitive. It is antisymmetric, and adjoining a least and greatest face gives a ranked partially ordered set. The verified geometry property gives its full-length maximal chains; thinness gives the diamond condition, and residue connectivity gives strong flag connectivity. These are exactly the abstract polytope axioms. The forty-eight subgroup-product inclusions are independently checked by literal set products. No string reconstruction theorem is needed for these finite rows.

Local residue identifications are marked isomorphisms. In (1), $\langle x, y\rangle$ is the exact torus group

$$\langle x, y \mid x^3, y^3, (x^{-1}y)^3, (xy^{-1}x)^a(xy)^b\rangle$$

of order $3(a^2 + ab + b^2)$. Its presentation maps onto the actual parabolic, and independently computed equal orders prove injectivity. The two designated spherical parabolics are checked in exactly the same way against $\langle s, t \mid s^3, t^p, (s^{-1}t)^2\rangle$. For (2) the facet group has presentation $\langle u, v \mid u^6, v^3, (uv)^2, (vu^{-2})^a(v^{-1}u^2)^b\rangle$, of order $6(a^2 + ab + b^2)$; the vertex group is the spherical

$\langle v, w \mid v^3, w^p, (vw)^2 \rangle$. The actual coset criteria already prove these groups describe residues; the local names are not inferred solely from an unmarked order match. □

The statement about chambers corrects the captions of both original tables. For example, Table 2 at $(p, a, b) = (3, 1, 2)$ has full automorphism group of order 168 and 336 chambers. A "flag" in the table caption must mean a chamber; even under that interpretation the printed g is not the number of flags in a chiral row.

3 The relation between the tables and the group names

The group of each Table 2 row embeds with index two in the corresponding Table 1 full group. This assertion is checked as a marked map, not by comparing the two orders. In the rotations (u, v, w) of (2), the map from (1) is

$$x \longmapsto v^{-1}u^{-2}, \qquad y \longmapsto v^{-1}, \qquad z \longmapsto w. \tag{3}$$

Every relator is satisfied, the image has half the order of Q, and the computed homomorphism is bijective onto that subgroup. Interchanging the first two images changes the orientation of the torus parameter; we retain (3) literally, including in the chiral rows. In the regular completions the corresponding involution map is

$$(r_0, r_1, r_2, r_3) \longmapsto (\rho_2, \rho_1^{\rho_0}, \rho_1, \rho_3).$$

The omitted generator ρ_0 is an involution outside this index-two subgroup. In the chiral cases an explicit involution outside the image is likewise retained. These data verify the split doubling assertions, including the unspecified actions in the source's semidirect-product notation. No converse classification is inferred from this construction.

The symmetric, projective linear, direct-product and real Coxeter names are checked by actual isomorphisms to their standard finite models. The real Coxeter models use their complete reflection presentations. The order-648 group is the monomial group $G(3, 3, 4)$:

$$\{(e_1, e_2, e_3, e_4) \in \mathbb{F}_3^4 : \sum e_i = 0\} \rtimes S_4,$$

acting on four blocks of three points. An explicit isomorphism to this model is retained. For the order-7680 group we use the named CHEVIE model G_{29} [4]. Its four roots are the rows of

$$\frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & -2 \\ -i & i & 1 & 1 \\ 1-i & -1+i & 0 & 0 \\ 0 & 1-i & -1+i & 0 \end{pmatrix}.$$

The reflections $v \mapsto v - 2\langle v, r\rangle r/\langle r, r\rangle$ act on the orbit of these roots. Exact arithmetic in $\mathbb{Q}(i)$ gives 160 roots and a permutation group of order 7680. Since the four roots span, that permutation action is faithful. The actual isomorphism to the Table 2 group is checked. These explicit models remove any reliance on the order alone for the source's abbreviations $[1\,1\,2]^3$ and $[1\,1\,2]^4$. The historical abbreviations are retained as labels; the displayed models specify the groups used here.

For $(p, a, b) = (3, 1, 4)$ the characteristic perfect residual has order 336 and is explicitly isomorphic to $\mathrm{SL}_2(7)$. We retain a subgroup A of order twelve, its isomorphism to A_4, and the equalities $A \cap N = 1$, $NA = G$. Thus the source's $\mathrm{SL}_2(7) \rtimes A_4$ is proved with its actual conjugation action. Its further extension in Table 1 is the split doubling above.

Proposition 2. *The group in Table 2 with $(p, a, b) = (3, 2, 2)$ is*

$$D = \{(s, t) \in S_5 \times S_4 : \mathrm{sgn}(s) = \mathrm{sgn}(t)\},$$

not the printed $A_5 \times S_4$.

Proof. On disjoint sets of five and four points, D is generated by

$$(1\,2\,3),\ (3\,4\,5),\ (6\,7\,8),\ (7\,8\,9),\ (1\,2)(6\,7).$$

These generate $A_5 \times A_4$ and one element of odd sign on each factor, hence the entire parity subgroup, of order 1440. The certificate gives a bijective homomorphism from the presented Table 2 group to this model.

The stable last term of the derived series of D is $A_5 \times 1$. Indeed the first terms are $A_5 \times A_4$, $A_5 \times V_4$ and $A_5 \times 1$. Its centralizer in D is $1 \times A_4$, of order twelve, since $C_{S_5}(A_5) = 1$. In contrast, the stable derived subgroup of $A_5 \times S_4$ is also $A_5 \times 1$, but its centralizer has order 24. This characteristic-subgroup invariant excludes an isomorphism. The finite checker verifies both derived series and centralizer orders. □

4 The three finite examples

Proposition 3. *Examples 4.4, 4.5 and 6.3 have all the group and incidence properties asserted in the source.*

Proof. For Example 4.4 put $u = ab$, $v = bc$, $t = b$, where a, b, c are the three printed involutions. Then t inverts u and v, and $(abc)^2 = 1$ is precisely $[u, v^{-1}] = 1$. Thus the presentation is $(C_3 \times C_3) \rtimes C_2$, with simultaneous inversion, of order 18. Conversely this model satisfies every defining relation, so it also supplies the lower bound. The three maximal parabolics have order six, and any two intersect in their common involution. They therefore have product of size 18. Every pair of cosets of different types is incident, giving $K_{3,3,3}$. All flags extend; there are 27 chambers, every panel has three completions, and every residue of rank at least two is connected. Its full type-preserving group is S_3^3. In particular the designated order-18 group is not flag transitive.

For Example 4.5 use the literal generators

$$(12)(34), \quad (26)(35), \quad (14)(23), \quad (14)(35).$$

They generate A_6 and satisfy all 256 marked subgroup intersections. Their maximal parabolics have orders $(24, 60, 60, 24)$ and the four type sets have sizes $(15, 6, 6, 15)$. The complete coset calculation gives 381 incidence edges, 1080 three-type flags and 720 chambers. All shorter flags extend. Among the 1080 panels, 720 have two completions and 360 have four. A residue of types $1, 2$ through a base flag of types $0, 3$ is $K_{4,4}$. A base flag of types $1, 2$ has a residue consisting of two disjoint 10-cycles. There are precisely 36 disconnected residues of rank at least two. Hence thinness, residual connectivity and designated flag transitivity all fail. The last assertion also follows from $|A_6| = 360 < 720$.

For Example 6.3 retain exactly the four permutations printed on page 14. Their group has order 72, with marked subgroup orders, in binary subset order,

$$1, 2, 2, 6, 2, 4, 8, 24, 2, 6, 6, 18, 4, 12, 24, 72.$$

All 256 intersections are correct. The rotation subgroup is the whole group. For $I = \{0, 1, 2\}$ and $J = \{1, 2, 3\}$,

$$|P_I| = |P_J| = 24, \qquad |P_I \cap P_J| = 8, \qquad |P_{I \cap J}| = 4.$$

The source's second involution ρ_1 belongs to the first intersection and not to $P_{I \cap J}$, giving a literal witness. These computations are independently repeated by closure of the permutation arrays and set intersections, without subgroup or coset routines from GAP. For Examples 4.4 and 4.5 this independent check also constructs every coset, every flag and every residue incidence graph. □

5 Scope, parameters and attribution

A minor domain qualification is needed in the source's rank-three toroidal discussion. Its unrestricted wording excludes $(1, 1)$, but one must also exclude the degenerate unit periods. For example,

$$\langle x, y \mid x^3, y^3, (x^{-1}y)^3, xy^{-1}x \rangle = \langle x \mid x^3 \rangle = C_3,$$

since $y = x^2$. Its three pair-difference parabolics are all the whole group; its standard coset system has one element of each type and is not thin. The usual admissible parameter domain is therefore necessary when invoking the prior toroidal classification. We make no claim here that a finite sampling proves that classification. Every period in Table 1 has norm at least three, and every Table 2 period has norm at least four; all are checked separately.

The source attributes the regular classification of locally toroidal polytopes to McMullen–Schulte, the rank-three chirality criterion to Breda D'Azevedo–Nedela, and the toroidal hypermap families to Corn–Singerman. Those classification statements remain explicit prior inputs wherever the general source recovery uses them. Our two finite tables are now verified positive examples independently of their completeness. Problem 9.2 and the unnumbered completeness conjecture remain separate questions. The old negative answers to Problems 9.3 and 9.4, including their restricted Lean scope, are acquired in the general recovery and receive no new credit here.

The source presentations, examples and positive conclusions belong to the source authors and their cited predecessors. The finite coset certificate theorem and its original implementation are reused from the accepted L003 recovery. The present contribution consists of the literal L052 applications, independent permutation and incidence checks, exact named-group maps, and the explicit corrections above. Named finite-group constructors, exact GAP presentation routines, and the named CHEVIE root model are declared computational and identification premises. They are not hidden inside a claim of Lean verification or new classification.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Math. 90 (2016), 1045–1067; accepted author manuscript, fixed 22-page edition.

[2] *Exact rotation presentations and coset certificates* and *Literal local periods and actual marked quotients*, accepted common proofs in the complete L003 recovery, Sections 00–01 of its shared supplement; exact editions and acceptance records acquired.

[3] *Chamber propagation and two regular orbits*, accepted Topic Chamber common manuscript, version 2.

[4] G. Malle and J. Michel, CHEVIE data for G_{29}, `src/tbl/cmplxg29.jl`; four literal root vectors, retrieved from the authors' software repository. See J. Michel, J. Algebra 435 (2015), 308–336, for the package.